

Experiment 9

Aim: Write a python program to perform lemmatization using NLTK.

Theory:

Lemmatization is a text normalization technique used in Natural Language Processing to reduce words to their base or dictionary form, known as the lemma. Unlike stemming, lemmatization considers the context and grammatical meaning of a word before converting it. For example, words such as running, runs, and ran are reduced to the base form run, while better may be reduced to good depending on context.

In NLTK, lemmatization is performed using the WordNetLemmatizer, which relies on the WordNet lexical database. The input text is first preprocessed by tokenizing it into individual words. Each token is then passed to the lemmatizer, which determines the correct base form using vocabulary and morphological analysis. Providing Part-of-Speech (POS) tags improves accuracy, since a word may have different lemmas depending on whether it is used as a noun, verb, adjective, or adverb.

Lemmatization helps reduce variations of words and improves consistency in text data. It is commonly used in applications such as information retrieval, text mining, sentiment analysis, and document classification, where treating similar word forms as a single term improves analysis results.

Source Code and Outputs:

```
!pip install nltk

import nltk
from nltk.stem import WordNetLemmatizer
from nltk.tokenize import word_tokenize

... Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('wordnet')
nltk.download('omw-1.4')
nltk.download('averaged_perceptron_tagger_eng')

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
[nltk_data] Package omw-1.4 is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger_eng.zip.
True
```

```
text = "The children are playing with better toys and running faster"

words = word_tokenize(text)

print("Original Words:", words)

Original Words: ['The', 'children', 'are', 'playing', 'with', 'better', 'toys', 'and', 'running', 'faster']

lemmatizer = WordNetLemmatizer()

lemmatized_words = [lemmatizer.lemmatize(word) for word in words]

print("Lemmatized Words (Default):")
print(lemmatized_words)

Lemmatized Words (Default):
['The', 'child', 'are', 'playing', 'with', 'better', 'toy', 'and', 'running', 'faster']
```

```
! nlk.download('averaged_perceptron_tagger')

from nltk.corpus import wordnet

# Function to map POS tag
def get_wordnet_pos(tag):
    if tag.startswith('J'):
        return wordnet.ADJ
    elif tag.startswith('V'):
        return wordnet.VERB
    elif tag.startswith('N'):
        return wordnet.NOUN
    elif tag.startswith('R'):
        return wordnet.ADV
    else:
        return wordnet.NOUN

# POS tagging
pos_tags = nltk.pos_tag(words)

lemmatized_words_pos = []

for word, tag in pos_tags:
    wn_tag = get_wordnet_pos(tag)
    lemmatized_words_pos.append(lemmatizer.lemmatize(word, pos=wn_tag))

print("Lemmatized Words (With POS):")
print(lemmatized_words_pos)

... Lemmatized Words (With POS):
['The', 'child', 'be', 'play', 'with', 'good', 'toy', 'and', 'running', 'faster']
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] /root/nltk_data...
[nltk_data] Package averaged_perceptron_tagger is already up-to-
[nltk_data] date!
```

```
for i in range(len(words)):
    print(f"{words[i]} --> {lemmatized_words_pos[i]}")

The --> The
children --> child
are --> be
playing --> play
with --> with
better --> good
toys --> toy
and --> and
running --> running
faster --> faster
```

Learning Outcomes:

I am able to write Python programs using NLTK to perform lemmatization by reducing inflected and derived words to their base forms using linguistic rules and vocabulary, thereby improving text normalization and supporting more effective downstream NLP tasks such as classification and clustering.